Camera-41_exp v1.0.1

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Why use Camera-41_exp?

Camera-41_exp helps set the digital camera exposure when digitising C-41 colour and black-and-white (B&W) film negatives with a VALOI setup. At a fixed aperture, it compares RAW captures of clear film base plus fog to find a shutter speed that gives a strong signal with room before clipping. Too much exposure can clip a sensor channel. Too little leaves a weak signal, making noise more visible when the scan is brightened.

Workflow

1. Open Camera-41_exp from Applications.
2. Use Add FB+F RAWs… to load an exposure bracket of the same empty film sample.
3. Enter Light setup and Film base + fog. Tick Same FB+F sample in all RAWs.
4. Press Analyse RAW images and read Results and Exposure comparison.
5. Verify the selected time with a fresh capture.

Measurement area and results

The app normally uses the central 70% of image width and height. Unexpected structure triggers a yellow rectangle for inspection and manual adjustment. Preview brightness does not affect measurements. PASS: at least 1/3 EV margin in all Bayer channels and no failing fraction at/above the limit. LOW: weak channel signal. REVIEW: inspect the channel results. FAIL: too many values reach or exceed the limit. Read the channel report and check the film sample and area before changing shutter time.

Independent tool

Camera-41_exp has no software or file-transfer connection to PRE or NEG and exports no JSON to them. If using PRE, set the chosen time, capture the current film's FB+F and two matched dark frames, then load those directly in PRE. Noise and S/N are measured there, not in Camera-41_exp.

Installation, startup and log

Open Camera-41_exp-1.0.1-python-macos.dmg. Drag the Apple Silicon app to Applications on an M-series Mac, or choose the Intel app for an Intel Mac. Add your own film-base RAWs with Add FB+F RAWs…. Open log shows the local diagnostic log. Quit the app before replacing it.

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https://github.com/chto0703009/Camera-41_exp

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